3.8 Quadratic Graphs

Introduction



Fig. 3.15

The trajectory followed by an object (say, a ball) thrown upward at an angle gives a curve known as a parabola. Trajectory of water jets in a fountain or of a bouncing ball results in a parabolic path. A parabola represents a Quadratic function.

A quadratic function has the form $f(x) = ax^2 + bx + c$, where $a,\ b,\ c$ are constants, and $a \neq 0$.

Many quadratic functions can be graphed easily by hand using the techniques of stretching/shrinking and shifting the parabola $y = x^2$ (We can easily sketch the curve $y = x^2$ by preparing a table of values and plotting the ordered pairs).

The "basic" parabola, $y = x^2$, looks like this Fig.3.16.

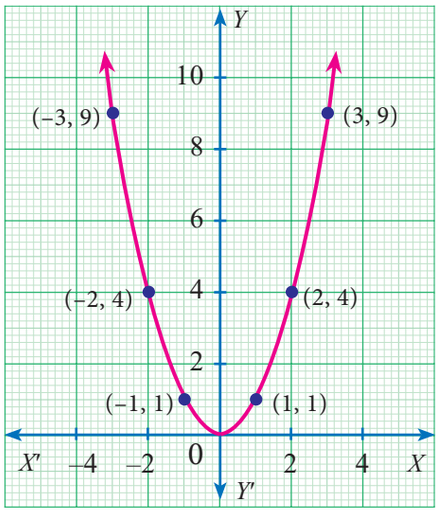


Fig. 3.16

The coefficient a in the general equation is responsible for parabolas to open upward or downward and vary in "width" ("wider" or "skinnier"), but they all have the same basic "∪" shape.

The greater the quadratic coefficient of x^2, the narrower is the parabola.

The lesser the quadratic coefficient of x^2, the wider is the parabola.

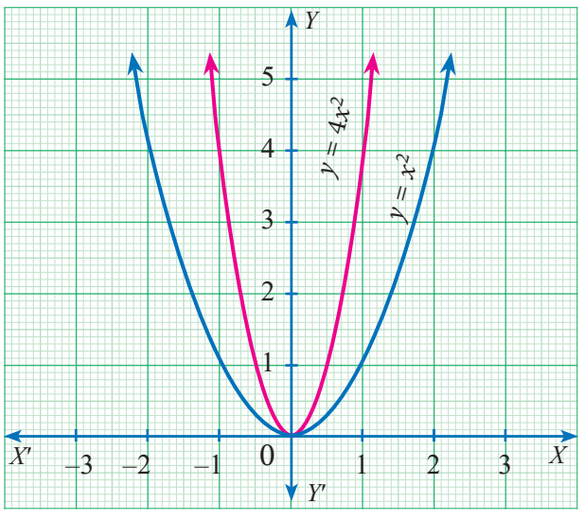


Fig. 3.17

Graph $y = x^2$ is broader than graph $y = 4x^2$

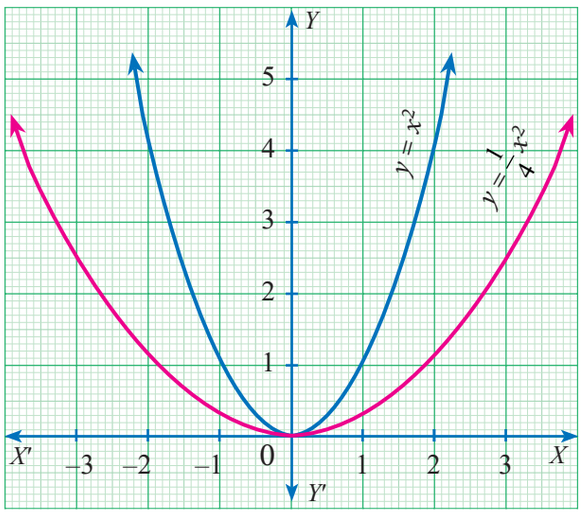


Fig. 3.18

Graph $y = x^2$ is narrower than graph $y = \frac{1}{4}x^2$

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